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PYTHON PROJECT-BASED LEARNING REPORT (WEEK 3)

1. Basic Information

Title of the Project:	Employee Management System & Library Management System
Programs Covered:	Program 1: Employee Management System Program 2: Library Management System
Name(s) of Student(s):	Soumyadwip Das (BWU/BTS/25/169) Ankita Paul (BWU/BTS/25/180) Sonia Naskar (BWU/BTS/25/195) Joyeta Mondal (BWU/BTS/25/214) Sayantan Bharati (BWU/BTS/25/503)
Programme & Semester:	B.Tech CSE & 3rd Semester
Course Name & Code:	Python Programming Lab (BES09013)
Name of the Guide/Supervisor:	MR. PRANAB GHARAI & MR. INDRANIL DAS
Project Date	24/07/2026





2. Introduction and Background

Week 6 covers two console-based CRUD systems: an **Employee Management System** and a **Library Management System**. Both use dictionaries, validation, unique IDs, and menu-driven operations.

3. Objectives of the Project (Week 6)

Program 1 - Employee Management System

- Create a console-based employee management system.
- Generate unique employee IDs using random.
- Validate employee name, department, and salary.
- Implement add, update, delete, search, and list operations.
- Test valid and invalid inputs.

Program 2 - Library Management System

- Manage members, books, and issue/return records.
- Generate unique IDs for members, books, and issues.
- Validate member email and phone details.
- Implement member and book management.
- Prevent double-issuing books and deleting members with active loans.

4. Problem Statement / Driving Question

Program 1: How can employee records be managed with unique IDs and valid inputs?

Program 2: How can members, books, and loans be managed while preventing invalid operations?



5. Methodology (Week 6 Activities)

Program 1 - Employee Management System

- Reused box-drawing functions for the interface.
- Created a random, collision-free employee ID generator.
- Added validation for name, department, and salary.
- Implemented add, update, delete, search, and list functions.
- Tested invalid department and salary inputs.

Program 2 - Library Management System

- Reused the boxed interface and menu structure.
- Created a shared unique-ID generator for members, books, and issues.
- Added validation for member name, email, and phone.
- Implemented member and book management.
- Used a separate ISSUED dictionary for loan records.
- Prevented double-issuing books and deleting members with active loans.

6. Expected Outcomes / Deliverables (Week 6)

- Working Employee Management System with CRUD operations.
- Working Library Management System with member, book, and loan management.
- Validated input for required fields.
- Safeguards against invalid library operations.
- Consistent boxed console interfaces.
- Future improvements: file/database storage and due-date tracking.



Program 1 - Unique ID Generation and Record Deletion

- ```
def generate_id():
 """Generate a unique 4-digit employee ID, e.g. E1234."""
 while True:
 emp_id = "E" + str(random.randint(1000, 9999))
 if emp_id not in EMPLOYEES:
 return emp_id # retry until a unique ID is found

def delete_employee(emp_id):
 if emp_id not in EMPLOYEES:
 print_boxed(["Employee not found."])
 return
 name = EMPLOYEES[emp_id]['name']
 del EMPLOYEES[emp_id] # remove the record from memory
```

[illegible]



## 8. Core Logic & Output (Program 2)

### Program 2 - Shared ID Generation and Issue Records

- generate\_id(prefix) creates unique IDs for members, books, and issues.
- issue\_book() prevents duplicate issues and creates a separate loan record.

```
def generate_id(prefix):
 """Generate a unique ID like B1234, M5678, I9012."""
 while True:
 new_id = prefix + str(random.randint(1000, 9999))
 if new_id not in BOOKS and new_id not in MEMBERS and new_id not in ISSUED:
 return new_id # unique across all three record types

def issue_book():
 book_id = input("> Enter book ID: ").strip().upper()
 if any(r["book_id"] == book_id for r in ISSUED.values()):
 print_boxed(["ERROR: Book is already issued."])
 return
 member_id = input("> Enter member ID: ").strip().upper()
 issue_id = generate_id("I")
 ISSUED[issue_id] = {"book_id": book_id, "member": member_id} # new issue record
```

### Program 2 - Library Management System

```
LIBRARY
MANAGEMENT
SYSTEM

Library Management System

--- MEMBERS ---
[1] Add a member
[2] Update a member
[3] Delete a member
[4] See all members
--- BOOKS ---
[5] Add a book
[6] Search a book
[7] See all books
[8] Issue a book
[9] Return a book
[10] See issued books
[0] Exit

> Enter option: 1

> Member name: ran
> Member email: ran@gmail.com
> Member phone: 1234567890
> Member student ID: 234

Member added successfully! ID: M7093

--- MEMBERS ---
[1] Add a member
[2] Update a member
[3] Delete a member
[4] See all members
```



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## 10. Tools & Technologies Used

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| Tool / Technology             | Purpose                                                                                      |
|-------------------------------|----------------------------------------------------------------------------------------------|
| Python 3.x                    | Core programming language used to implement both console systems                             |
| Python Dictionaries           | In-memory storage of employees, members, books, and issue records, each keyed by a unique ID |
| random module                 | Generating unique, prefix-based IDs for employees, members, books, and issue records         |
| Input Validation Loops        | Re-prompting until a valid department/salary, or a valid member email/phone, is provided     |
| String Formatting (f-strings) | Rendering aligned tabular listings of employees, members, books, and issued loans            |
| Functions & Control Flow      | Modularising add/update/delete/search/issue/return operations across both systems            |

## 11. References

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- Python Official Documentation: <https://docs.python.org/3/>
- Python random module Reference: <https://docs.python.org/3/library/random.html>
- Python Dictionaries - Data Structures Guide: <https://docs.python.org/3/tutorial/datastructures.html>
- Brainware University - Machine Learning Course Guidelines



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## **12. Signatures**

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**Signature of Student(s):**

**Date of Submission:**

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**Signature of Guide/Supervisor:**

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